Jack B. Hughes

(720) 393-9096 — hughe307@purdue.edu — LinkedIn — West Lafayette, IN

Experienced designer seeking fall internship in aerospace mechanical design.

EDUCATION

Purdue University School of Aeronautics and Astronautics *(West Lafayette, IN)*

B.S. in Aeronautical and Astronautical Engineering (major) Expected 5/2027

Computer Science (minor) and Mathematics (minor)

SKILLS AND PROFICIENCIES

Computer Aided Design: Autodesk Inventor (4000+ hrs), Siemens NX (150 hrs), Onshape (100 hrs)

Programming: C/C++ (1600+ hrs), Java (1500+ hrs), Python (200 hrs), MATLAB (200 hrs), HTML/CSS/JS (100 hrs)

Other: Additive Manufacturing Design & Iteration (5 years), Rapid Prototyping (9 years), Project Management (7 years)

DESIGN & LEADERSHIP EXPERIENCE

Purdue University ACM SIGBots Robotics Team *(West Lafayette, IN)* 08/2022–05/2026

Competition Team Lead/Chief Engineer (2022-2026)

- Served as project manager and chief engineer of a 12+ person VEXU robotics team that competed globally against other universities in head-to-head matches with both operator and autonomously controlled periods
- Applied Autodesk Inventor to design 8+ robots over four years and created 1000+ custom parts designed for FDM 3D printing, resin 3D printing, CO2 plastic laser cutting, Fiber metal laser cutting, CNC machining, and brake forming
- Developed, iterated, and maintained a custom chassis tracking & movement library in C/C++ used across all seasons
- Architected multiple integrated software systems, including a computer vision-aimed turret robot: Photos, GitHub

President (2025-2026), Treasurer (2024-2025)

- Managed 100 members across four competition teams, collaborating with university and corporate stakeholders
- Led the club to another world championship victory in 2026 by managing $30,000 annual budget and implementing new recruitment, training, and team-building programs, improving retention rates by 300%

PROFESSIONAL ENGINEERING EXPERIENCE

The Aerospace Corporation *(El Segundo, CA)* 05/2026–Present

Engineering Intern IV – Space Architecture Concepts

- Designed, manufactured, and programmed a debris removal spacecraft by developing and iterating on a novel low-mass space robotics technology, autodynamic flexible circuit (AFC) actuators
- Contributed to a multidisciplinary concept design study as the structures/CAD lead using Siemens NX
- Created and maintained CAMEO (UAF) models to improve understanding of ever-changing USSF organizational structure and capability procurement processes
- Modeled and analyzed various spacecraft constellation architectures for resiliency

Sierra Space *(Centennial, CO)* 05/2024–08/2025

Technical Intern – Sierra Defense Systems Engineering

- Designed and manufactured a 3D-printed scale model of a satellite for use in classified meetings with customers
- Diagrammed and architected system interfaces of early-stage satellite programs in CAMEO
- Developed a CONOPs for a satellite program, prepared technical deliverables, and presented to customer stakeholders
- Implemented this CONOPs as a set of CAMEO simulations supporting an interactive full-system behavioral model
- Designed a parametric framework for spacecraft power consumption modeling in cooperation with responsible engineers
- Created and owned the company's first master test plan document, dictating program test strategy from the system level down to every component test in collaboration with the customer, subject matter experts, and program management
- Developed and implemented a novel technique for modeling and simulating spacecraft subsystem control loops and algorithms in CAMEO to coordinate across disciplines and optimize workflows